7. The Agentic Office

How Microsoft, Google, and AI agents reorder knowledge work

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1. Executive Summary

Office software — the integrated suite of document, spreadsheet, and presentation tools plus the communication layer of email, chat, and video — is the broadest horizontal software category in the enterprise: its user base spans essentially the entire knowledge workforce, it claims the largest share of knowledge-worker attention of any application layer, and it sits at the terminal point of most workflows in the organization. These three structural properties make the category, dominated by the Microsoft 365 / Google Workspace duopoly, one of the largest revenue pools in software — Microsoft 365 alone generated roughly $95 billion in FY2025 — and the single largest lever for any technology that claims to raise knowledge-work productivity.

Agentic AI is that technology, and 2025–2026 marks the moment its entire competitive field — incumbents, frontier model labs, and AI-native startups alike — converged on the office layer as its first and decisive battleground. This report traces how the category arrived here and what happens next, organized around four analytical devices.

The triad framework. Office software has always been the institutionalized form of three permanent cognitive operations of knowledge work: qualitative reasoning (the document), quantitative analysis (the spreadsheet), and visual persuasion (the presentation). For forty-five years, software improved each axis individually while the labor of integrating them — moving numbers into narratives into slides — remained human. The defining act of the agentic era is the transfer of that integration labor to software: agents treat the three axes not as separate workspaces but as rendering targets of a single reasoning process. Value migrates upstream accordingly, from the apps and their files to the organizational context layer and the agent harness that orchestrates the pipeline.

The arena framework. The competition unfolds on terrain owned by two estates with opposite architectures: Microsoft's app-native container model (files exist, the reference renderer rules) and Google's file-less service model (no files, only database records behind an API gate). In 2026 both opened their estates to agents through API, MCP, and CLI surfaces — but with opposite postures. Google, the challenger, commoditizes access (an open-source CLI, a public MCP server, explicit hospitality to third-party agents). Microsoft, the incumbent, meters it (Work IQ context sold through Copilot Credits inside the tenant trust boundary). The pattern is historically familiar: the challenger opens, the incumbent gates. What both protect identically is not functionality but context — the semantic model of how the organization works, which is the scarce input that caps agent quality.

The relocation of lock-in. File formats, the original moat of this industry, are strategically spent. OOXML is a neutralized output codec that every agent generates through open libraries; Markdown is becoming the agents' working lingua franca precisely because no one can own it; spreadsheet and presentation logic is being decomposed into data plus code plus web rendering, dissolving the fused proprietary container. Lock-in does not disappear — it relocates to protocols (MCP, A2A, the emerging commerce protocols), execution sandboxes, verification interfaces, and above all the organizational context graph.

The knowledge-architecture question. Beneath the contest of apps and agents runs a deeper transition in what enterprise knowledge *is*. The agent-native working stack now has a concrete, partially standardized form — Markdown for qualitative knowledge (formalized in June 2026 by Google's Open Knowledge Format), columnar Parquet for quantitative data, JSON state plus web rendering for the visual layer — and its maturation points toward the enterprise **knowledge network**: the organization's knowledge as a curated, linked, machine-executable estate, maintained largely by agents, from which documents, models, and decks are compiled on demand. This report treats the knowledge network as the strategic endpoint of the office category, frames the asset it creates as **knowledge capital**, and analyzes it through three explicit scenarios rather than a single forecast. Google's Gemini Notebook (rebranded from NotebookLM in July 2026) — examined here as the duopoly's first shipped *source-first* office surface, one that generates documents, tables, slides, and video as renderings of a grounded corpus — is read as the working prototype of this end-state, and as the second front of a two-surface Google strategy that Microsoft's single-surface, app-preserving defense cannot symmetrically copy.

The report projects three phases: an Actuator Era (now–2027) in which agents operate the existing apps; a Rendering Era (toward 2030) in which files become on-demand views of a living context substrate and per-seat pricing gives way to hybrid seat-plus-consumption models — a transition already visible in Notion Credits, OpenAI's agent pricing, and Microsoft's metered consumption; and a Human-Boundary Contraction in which the triad's addressable market is redefined by the shrinking surface where human attention intervenes. The three axes will not decay uniformly: presentation production commoditizes fastest, the spreadsheet endures longest as the audit surface for agent work — for as long as verification itself remains a human act — and the document splits into a dissolving communication function and a persisting record function.

For investors, the central conclusion is that the agentic productivity war and the contest for the next form of the office layer are the same war. Microsoft has mounted the strongest incumbent defense in the industry's history — a fourth bundling campaign (E7), a context monopoly play (Work IQ), and a hedged multi-model supply chain that places its sharpest insurgent, Anthropic, inside its own product. Google holds the architectural high ground (it built the file-less, API-gated model two decades early) and is playing offense through radical openness. The harness layer has been repriced by the market itself — Meta's $2B+ acquisition of Manus, a company with no proprietary frontier model, is the cleanest single datapoint that scarcity is migrating from model weights to orchestration. Two positions are, in this report's judgment, mispriced by the prevailing framing: Microsoft's ownership of GitHub, VS Code, and the Windows execution layer makes it the owner of the open knowledge stack's natural habitat — a hedge on the disruption of its own Office franchise that the format-war narrative ignores — and Gemini Notebook constitutes unpriced optionality inside Alphabet on the source-first office, acquired bottom-up at consumer scale before the enterprise contest begins. The risks worth pricing are governance failure (prompt injection at organizational scale), the geopolitical capture of agent technology (China's blocking of the Manus acquisition), the possibility that bundling, for the first time in four campaigns, attracts decisive regulatory interference, the revenue valley of the seat-to-consumption transition, and the semantic lock-in the knowledge network reintroduces one level above the formats it frees.

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2. The Universal Layer of Knowledge Work

2.1 The Broadest Horizontal Software: A User Base Coextensive with the Knowledge Workforce

Nearly every category of enterprise software is vertical in the functional sense: it serves a department, a role, a process. CRM belongs to sales, ERP to finance and operations, HRIS to human resources, design tools to designers, observability stacks to engineers. Each addresses a subset — often a small one — of the organization's knowledge workers.

Office software is the conspicuous exception. Its user base is not a subset of the knowledge workforce; it *is* the knowledge workforce. A salesperson, an accountant, an engineer, and a chief executive perform different jobs, but all of them write documents, build tables, present arguments, and communicate through email, chat, and meetings. Whatever the role, the common operations of knowledge work — qualitative description, quantitative analysis, visual persuasion, and communication — are identical, and the office suite is the institutionalized form of those common operations.

The horizontality shows up immediately in scale, though the two estates' headline figures measure different things. Google says its Workspace applications are used by more than three billion people — a figure that includes free consumer accounts — while Microsoft 365 counts more than four hundred million paid commercial seats. The two numbers are not directly comparable, but together they demonstrate that the two estates reach an exceptionally large share of the world's knowledge workers — a population on the order of a billion people. Among work-specific software, no other category approaches this penetration of its addressable population.

2.2 Time-Share Dominance: The Largest Claim on the Working Day

Penetration understates the position; the deeper measure is residence time. Microsoft's Work Trend Index telemetry shows that, within Microsoft 365, the average worker spends 57 percent of application time communicating — email, meetings, chat — and 43 percent creating documents, spreadsheets, and presentations. The measurement does not directly compare Office against every other business application, but it demonstrates how much of the knowledge-work day is mediated by the office and communication layer — for most knowledge workers, plainly more than any other single software environment claims.

Time share matters because it is the denominator of every productivity calculation. A technology that improves a tool used two hours a week competes for a sliver of output; a technology that improves the substrate of the entire working day competes for the whole of it. This arithmetic returns with force in Chapter 4.

2.3 The Last Mile: Where Every Workflow Converges and Every Decision Is Made

The third structural property is positional. Office software is the terminal segment — the last mile — into which the outputs of every other enterprise system flow. The ERP's numbers are extracted into Excel to be analyzed; the CRM's pipeline becomes the quarterly-review slide deck; the data warehouse's query results are pasted into a memo; and the conclusions of every system are ultimately argued in email threads and decided in meetings. The industry aphorism that "all roads lead to Excel" names a real topology: vertical systems *store* the organization's data, but the organization's actual reasoning and deciding are *performed* in the office layer.

This position is why office software cannot be displaced by any combination of vertical tools. A company can in principle replace its CRM, its ERP, or its ticketing system one by one; it cannot replace the layer in which the outputs of all of them are synthesized, contested, and turned into decisions, because that layer is not a function — it is the medium of organizational cognition itself.

2.4 The By-Product Asset: The Densest Repository of Organizational Context

Because everything passes through the office layer, the office estate accumulates — as an automatic by-product of use — the densest available record of how the organization actually works: who discussed what with whom, which document fed which meeting, which analysis preceded which decision. Emails, files, calendars, chat threads, and meeting transcripts together constitute an implicit graph of the organization's structure, priorities, and reasoning history.

For three decades this by-product was inert, valuable mainly for compliance and discovery. The agentic era converts it into the category's most strategic asset, because — as Chapter 4 will argue — the quality of an AI agent's work is capped not by its model but by its context, and no other system in the enterprise holds context of comparable density. Microsoft's Work IQ and Google's Workspace Intelligence, both examined in Chapter 5, are nothing more or less than the productization of this by-product.

2.5 Economic Expression: One of Software's Largest Revenue Pools, Held by Two Players

The economic expression of horizontality, time-share dominance, and last-mile position is a market of extraordinary concentration and weight. Microsoft 365's Commercial and Consumer lines generated approximately $95 billion in FY2025 — one of the largest recurring revenue streams in the history of the software industry — before adding Google Workspace's undisclosed revenue; combined, the duopoly's office revenues rival the sum of dozens of vertical SaaS categories. The category's strategic importance derives not from equaling the rest of enterprise software in revenue, but from its reach across every role, its claim on knowledge-worker attention, and its position at the terminal point of workflows. The duopoly structure is itself evidence of the category's nature: as the historical chapters will show, repeated waves of challengers were absorbed or repelled because the category is purchased not as a set of features but as the basic infrastructure of knowledge work — and infrastructure markets consolidate.

This is the structure within which everything that follows takes place. The remainder of this report treats Microsoft 365 and Google Workspace not merely as two competitors but as the two *estates* — territories of users, data, and institutional habit — upon which the entire agentic AI field, the estate owners included, must now maneuver.

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3. The Triad and Its Three Eras

3.1 The Triad Framework

The essence of office software has been constant since its origin: it is the tooling for three permanent cognitive modes of knowledge work — the narration of qualitative information (the **Document**), the manipulation of quantitative data (the **Spreadsheet**), and the visual delivery of persuasion (the **Presentation**) — joined, from the 1990s onward, by a fourth layer of **communication** (email, then chat and video). The industry's history is best read as the history of a single question: *at what depth are these three axes integrated, and by whom?* The answer to that question defines each era, and its final transformation defines the agentic future.

3.2 Stage 1 — The Standalone Era (1979–1990): Three Axes Born Apart, Integrated by Humans

The three axes were invented separately, by separate companies. WordStar and then WordPerfect opened the document axis; VisiCalc and then Lotus 1-2-3 opened the spreadsheet axis; Harvard Graphics and the Macintosh-born, Microsoft-acquired PowerPoint opened presentation. Microsoft was still a peripheral applications player, but its early investment in graphical platforms proved decisive: Word originated on Xenix and MS-DOS in 1983, Excel debuted on the Macintosh in 1985, and that early GUI experience prepared Microsoft's applications for the platform transition to come.

The defining feature of Stage 1 is that integration across the axes was zero. Numbers computed in a spreadsheet were retyped by hand into reports; charts were printed and physically pasted into slides. The pipeline that constitutes actual knowledge work — analyze, narrate, persuade — existed entirely inside human hands and heads. Software raised the productivity of each axis individually; the integration labor between them was the human's job, and would remain so for forty-five years.

3.3 Stage 2 — GUI and Suite Bundling (1990–2005): The Birth of Office Hegemony

Microsoft's victory rested on controlling the platform transition it forced. As Windows displaced DOS, Microsoft pre-optimized Word, Excel, and PowerPoint for the graphical interface while the DOS-era category leaders hesitated, then in 1990 fused them into a single product: Microsoft Office. The integration was commercial (one SKU, one price), interface-level (a common menu and shortcut grammar), and — thinly — data-level (OLE embedding). Measured against the pipeline it remained shallow: three apps, three artifacts, three formats, with the human still orchestrating the flow between them. Strategically, however, shallow integration sufficed. The standalone champions discovered that excellence on one axis cannot outsell adequacy on three bundled axes at a bundle price, and Stage 2 established the industry's first iron law, against which every later challenger would collide: **the three axes are not sold separately.** By the late 1990s Microsoft's share exceeded ninety percent, the .doc/.xls formats had become the de facto interchange standard of business itself — the network-effect moat examined in Chapter 7 — and Exchange/Outlook extended the bundle into the communication layer.

A parallel track deserves note: Lotus Notes (1989) was an early and commercially successful attempt to move work from document-centric applications toward networked collaboration. It ultimately lost the mass-market communication layer to Exchange/Outlook and later web-based tools, but its scale demonstrated that collaboration could itself become a software platform — terrain that Slack would attack two decades later, and that agents would attack two decades after that.

3.4 Stage 3 — Cloud and SaaS (2006–2022): Google's Challenge and the Settling of the Duopoly

Google's acquisition of Writely and launch of Google Docs in 2006 introduced two simultaneous ruptures. Technically, the document ceased to be a file and became a living object at a URL — real-time co-editing, granular version history — and the basis of lock-in began migrating from file formats to the sharing graph: who has access to what. Commercially, perpetual licenses gave way to subscription; Microsoft's defensive conversion to Office 365 became one of the most successful business-model transitions in software history, replacing episodic upgrade revenue with predictable per-seat ARR. Notably, Google never functionally displaced Office in the large enterprise; its role was to change the rules of evaluation — collaboration became a mandatory axis — and the market settled into the duopoly that frames this report.

Stage 3's most under-appreciated fact, however, is what did *not* change: the triad structure itself. Google reproduced the identical three-app division — Docs, Sheets, Slides — in the browser. The cloud changed where files live and who edits them; it left untouched the forty-year-old partition of knowledge work into a document app, a table app, and a slide app. Serious challenges to the partition came only from the niche — Notion dissolving the document/database boundary, Airtable the spreadsheet/database boundary, Canva attacking presentation obliquely through design — proving the partition a legacy of 1980s software architecture rather than a necessity of human cognition, but never moving the enterprise standard. The era also taught the unbundling lesson: single-function challengers are either reabsorbed by the bundle (Slack, blunted by Teams' free bundling and sold to Salesforce) or survive only by creating a new category (Notion's work-OS, Canva's design platform).

3.5 The 45-Year Invariant — and the Copilot Interlude (2023–2025)

Across three eras, two things never changed: the triad's partition, and the assignment of integration labor to the human. Real knowledge work is one continuous pipeline — analyze the data (quantitative), narrate its meaning (qualitative), persuade the audience (visual) — that software artificially split into three applications, with the splice costs (re-keying numbers, refreshing charts, synchronizing versions) paid in human time.

The generative-AI Copilot wave of 2023–2025 was, structurally, an extension of Stage 3 rather than a break from it. AI was inserted into Word, into Excel, into PowerPoint — separately, as a feature of each app, priced as an add-on atop the per-seat model. GitHub Copilot proved the concept, ChatGPT detonated the demand, and Microsoft 365 Copilot ($30 per seat per month) and Google's Workspace AI institutionalized it. But the paradigm remained human-in-the-loop — the human drives, the AI assists — and the triad's partition, the human's integration role, and the per-seat economics all survived intact.

That is precisely the configuration the agentic era attacks. The point of attack is not any one axis. It is the invariant itself.

4. Why the War Converges on Office

If Chapter 2 established that office software is the universal layer of knowledge work, this chapter establishes the forward-looking corollary: in the agentic era, no other software category offers a comparable lever on knowledge-work productivity — which is why the entire competitive field has converged on it. The argument has five layers.

4.1 The Arithmetic of Productivity: Gains = Improvement Rate × Coverage

The total productivity contribution of any AI deployment is the product of two terms: how much it improves a task, and how much of total work that task represents. A vertical agent that makes a specialized workflow ten times faster, where that workflow occupies two percent of the organization's knowledge-work hours, moves the aggregate by a fraction. Office and communication work — per Section 2.2 — claims the largest share of the software-mediated working day. At equal improvement rates, the agent that automates the universal layer produces the largest total effect by construction. There is no second lever of comparable length: any technology that intends to move the macroeconomic variable called "knowledge-work productivity" must pass through the office layer, because that is where the hours are.

4.2 Delegation Requires the Substrate: Agents Must Work Where Work Lives

Agentic AI's core promise is delegation — not assisting a task but receiving it whole. Delegation has a precondition: the agent must have access to the substrate in which the work exists. For knowledge work, that substrate is precisely the triad's artifacts and the communication stream — the documents, workbooks, decks, inboxes, calendars, and threads of Chapter 2.

This is why, in 2025–2026, nearly every major player's agent strategy chose office as its first battlefield. Microsoft embedded Agent Mode inside Word, Excel, and PowerPoint and built Copilot Cowork; Anthropic generalized Claude Code into Cowork and shipped add-ins directly inside the Microsoft apps; Google wired Gemini Spark natively into Gmail and Workspace; and the AI-native entrants — Skywork, Genspark, Manus — uniformly lead with document, spreadsheet, and slide generation as their core value proposition. The convergence is itself the market's verdict: the home territory of the agent era is the office layer.

4.3 The Context Ceiling: Agent Quality Is Capped by the Office Estate

The decisive determinant of agent output quality is not the model but the context: what the agent knows about this organization, this project, these people, this history. Invert the proposition and a strategic theorem follows — whoever holds the best context can field the best agents.

The largest store of organizational context is the office estate (Section 2.4). Microsoft's description of Work IQ makes the point explicitly: by continuously processing email, calendar, meetings, chats, files, people, and collaboration patterns, it claims to deliver "intelligence you cannot get from the data alone — a real-time model of how your organization operates." That sentence is a declaration that the raw material of the agent era is mined from office usage. The office layer is thus doubly central: it is the object of the agent's work and, simultaneously, the fuel that makes the agent competent.

4.4 The Interface Argument: Communication as the Human–Agent Channel, the Triad as Verification Surface

In the agentic era the communication layer acquires a new function: it becomes the interface through which humans work with agents. The design choices are telling. Google gives Spark a dedicated Gmail address — you delegate to it the way you email a colleague. Anthropic describes interacting with Cowork as "less like a chat back-and-forth and more like leaving messages for a coworker." The channels humans built to collaborate with humans — email, chat, meetings — are being repurposed, unmodified, as the channels for collaborating with agents.

Symmetrically, the triad's artifacts acquire a new function on the output side: they are the surfaces on which humans verify agent work. A human audits an agent's quantitative reasoning in a spreadsheet grid, reviews its narrative in a document, and judges its argument in a deck. The entrance of delegation (communication) and its exit (artifacts and review) are both office surfaces.

4.5 Form Dissolves, Function Persists: What "Office" Means After the App

The deepest layer of the argument guards against a misreading. This report will project (Chapter 8) that office *apps* are demoted to actuators and office *files* to rendered views. That is a forecast about the dissolution of the category's current product form — the app, the file, the per-seat license — not of its function. The functions office software has institutionalized — qualitative reasoning, quantitative analysis, persuasion, communication, and the recording of all four — are permanent constituents of organized human activity. Agents do not abolish these functions; they become their new performers.

The market for "office software in the agentic era" is therefore properly defined as the market for **owning the next form of the universal function layer of knowledge work** — and because of that universality, the value available to its winners is, by the logic of Chapter 2, among the largest prizes in enterprise software. The chapters that follow analyze the arena in which that ownership is being contested, and the players contesting it.

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5. The Arena

Before analyzing players, the terrain. The agentic contest is not fought on neutral ground: it unfolds on top of two installed estates — Microsoft 365 and Google Workspace — that hold the users, the artifacts, and the context. The estates differ in architecture, in how they have productized context, and in the posture of the gates they have opened to agents — and, on Google's side, a second, architecturally distinct surface has emerged alongside the estate itself: Gemini Notebook, the first shipped office product organized around sources rather than artifacts. These three differences, and that one asymmetry, define the physics of everything in Chapter 6.

5.1 Two Estates, Two Architectures: App-Native Container vs. File-less Service

Microsoft's architecture is the app-native container. Work materializes as files — OOXML containers (.docx/.xlsx/.pptx) — that exist independently of any service, can be carried away, and can be opened by any conforming application. The architecture's gravity (analyzed fully in Chapter 7) lies not in the format, which is formally an open standard, but in the reference-renderer position: interoperability in practice means "does it open correctly in Word/Excel," and the arbiter of that test is Microsoft.

Google's architecture is the file-less service. A Google Doc has no file format at all: the .gdoc on a synced drive is a few lines of JSON pointing to a document ID; the document itself exists only as a database record on Google's servers, stored as an accumulated sequence of edit operations (the architecture that enables real-time co-editing and infinitely granular version history). Since 2021 Google Docs has not even rendered through standard HTML: it draws the document onto a canvas, using the browser as a runtime for a proprietary rendering engine rather than as an interpreter of web-standard markup. Export to .docx or PDF is possible but lossy — comments, history, and collaboration state stay behind.

The strategic readings are mirror images. Microsoft's model has zero service lock-in at the file layer (you can leave with your files) but retains rendering gravity. Google's model has zero *format* lock-in — there is no format — and the strongest *service* lock-in of the three architectures this report distinguishes (Chapter 7 adds the third: the open web substrate). Google, in effect, implemented two decades early the end-state this report projects for the whole industry — *truth lives in a service-side graph; files are rendered views* — with the single difference that its graph is proprietary. The phrase "Google's format" is therefore a category error with analytical content: **Google's format is its API.**

5.2 The Context Layers: Work IQ vs. Workspace Intelligence

Both estates spent 2026 converting the by-product of Section 2.4 into a product.

Work IQ is Microsoft's name for the semantic layer built by continuously processing the tenant's email, calendar, meetings, chats, files, people, collaboration patterns, and line-of-business systems into a real-time model of the organization. Its APIs reach general availability on June 16, 2026, positioned explicitly as "the best way for agents to interact with Microsoft 365 data and apps," with three protocol faces — A2A for agent-to-agent delegation, REST for application integration, MCP for tool-style access — and with data, context, and agent actions kept inside the tenant trust boundary, auditable and admin-governed. There is no separate Work IQ SKU; usage is billed through Copilot Credits. Work IQ is, in short, the office by-product asset converted into a metered, governed, monetized input for agents.

Workspace Intelligence, announced at Cloud Next '26, is Google's counterpart: a real-time understanding of the user's work across Docs, Slides, Gmail, Drive, and Chat, feeding both Gemini's in-app features and the Spark agent. Google's productization currently trails Microsoft's in enterprise packaging — Work IQ ships with a formal API surface, billing meter, and governance frame — but Google's underlying position is architecturally privileged: because the estate was always file-less and API-gated, the context graph has been the native form of Google's data from the start rather than a layer extracted from files. (The same context ambition extends beyond Workspace on the Cloud side: Knowledge Catalog — the 2026 evolution of Dataplex — positions itself as a context engine for enterprise data at large, and Chapter 7 returns to the open knowledge format it introduced alongside.)

The context layer is where this report locates the principal lock-in of the agentic era. It is the one asset that cannot be replicated by a model lab or a startup, because it accrues only from holding the estate through which the organization's work flows.

5.3 The Gates: Agent Access Postures Compared — The Incumbent Meters, the Challenger Opens

In 2026 both estates formally opened agent-facing access surfaces across the same three modalities — API, MCP, CLI — confirming the actuator thesis: the suites are being re-presented as operable surfaces for machine workers. The postures, however, are opposite.

Google commoditizes access. At Cloud Next '26 it announced a Workspace agent-tools suite — a Workspace MCP server, a CLI, and remote MCP integrations — and opened the MCP server to public developer preview from May 1, 2026, exposing dedicated agent tools for Gmail (drafting, search, read/write) and Drive (fetching, permissions, listing, upload). The Workspace CLI ("gws," shipped early March 2026) is the sharper signal: open-source (Rust, Apache 2.0), a single interface to every Workspace API, structured JSON output designed for machine consumption, more than one hundred bundled agent skills packaged as SKILL.md files, an embedded MCP server option, and — most remarkably — explicit documentation for connecting *third-party* agents, including Anthropic's Claude Code and viral independent assistants, with the CLI pulling API specifications from Google's Discovery Service at runtime so its coverage auto-synchronizes with the full API estate. Google is not selling access; it is paving a public road into its estate and inviting every agent, friendly or rival, to drive on it. (One tempering note: simultaneously with the MCP preview, Google introduced usage tiering on Workspace APIs and MCP for new projects — openness at the on-ramp, metering at scale.)

Microsoft meters access. The equivalent stack exists — Work IQ APIs in three protocols, a governed catalog of Work IQ MCP servers manageable from the admin center, the Agent 365 SDK and CLI, Copilot Studio, the Agent Store — but every path runs through three gates: tenant-admin enablement, the Copilot licensing frame, and Copilot Credit metering. What Microsoft offers beyond raw functionality (which Graph APIs already provided) is the Work IQ context itself: richer input, governed environment, priced accordingly. Microsoft is not selling access either; it is selling *context with governance*, and gating access as the collection mechanism.

The asymmetry is not stylistic but positional. For the enterprise challenger, opening to agents is offense: if agents find one estate easier to operate, then as the agent-mediated share of work grows, that estate's relative attractiveness compounds — a new distribution channel. For the incumbent, indiscriminate openness would route around its own monetization (Copilot seats plus credits); gating is rational defense. The pattern is the same one that governed the format wars: the side under pressure opens (Microsoft opened OOXML under ODF and regulatory pressure in the 2000s; Google opens the agent gates now), the side in possession meters.

For a third-party agent today, the practical scorecard reads: Google offers materially more convenience and lower friction; Microsoft offers deeper organizational context at higher commercial and governance cost; functional breadth is roughly comparable but differently shaped (Google's coverage auto-syncs to its whole API estate; Microsoft exposes deeper in-app machinery — the Excel calculation engine, PowerPoint rendering — but largely inside the Copilot commercial frame).

5.4 Google's Second Surface: Gemini Notebook and the Source-First Office

The analysis so far has treated Google's position as one estate with one architecture. Since roughly late 2024, that has been incomplete. Alongside Workspace, Google has grown a second office surface whose organizing principle is the inverse of everything the triad institutionalized — and whose trajectory matters more to this report's endgame than any single Workspace feature.

What Gemini Notebook is. Gemini Notebook — renamed from NotebookLM on July 16, 2026, the current name used throughout this report — is a source-grounded working surface: the user (or team) assembles a corpus — PDFs, Google Docs and Slides, websites, YouTube transcripts, audio files, Markdown files, spreadsheets, EPUBs, with individual sources running to hundreds of thousands of words — and the system grounds everything it says in that corpus, with citations back to the exact passage. Around this grounding discipline Google has built a generation studio whose expansion tells the strategic story in miniature: podcast-style Audio Overviews (September 2024) made the product a consumer phenomenon; Video Overviews followed in 2025; slide decks and infographics arrived in November 2025 on the Nano Banana Pro image model; structured Data Tables, exportable to Google Sheets, arrived in December 2025; and by March 2026 the surface had acquired slide revision loops, cinematic video, PPTX export, and artifact generation directly from chat. As of mid-2026 the product runs on Gemini 3.5, integrates Deep Research for corpus-building, exposes notebooks as queryable context from within Gemini itself, and ships in three tiers — consumer, NotebookLM Plus through Workspace and Google One, and NotebookLM Enterprise on Google Cloud with IAM roles, notebook sharing, and residency controls. On July 16, 2026, Google renamed the product **Gemini Notebook** — retaining it as a standalone product while deepening synchronization with the Gemini app and announcing its extension into Search's AI Mode — and began rolling out a per-notebook "secure cloud computer" that writes and executes code natively against a notebook's sources, citing more than 30 million users and over 600,000 organizations at the rename. Adoption came bottom-up, through education, research, and consumer virality, before the enterprise packaging existed — the browser's adoption path, not the ERP's.

Three analytical readings follow.

First, Gemini Notebook inverts the office paradigm: artifact-first becomes source-first. Every triad application for forty-five years has opened onto an empty artifact — a blank document, an empty grid, a title slide — into which the human pours knowledge. Gemini Notebook opens onto a corpus, and every artifact is generated *from* it: the report, the data table, the deck, the podcast, and the video are alternative renderings of one grounded source set, produced on demand and regenerated at will. This is precisely the model this report projects for Phase B — files as compiled views of a living context substrate (Section 8.2) — shipping today as a mainstream product. A notebook is, in miniature, an enterprise knowledge network: a curated, bounded corpus with generated views. The reader should hold that sentence; Chapters 7 and 8 scale it up.

Second, Gemini Notebook demonstrates the triad's collapse in a single surface. One corpus yields the document (reports), the table (Data Tables flowing into Sheets), and the deck (slides flowing out as PPTX) — plus formats the triad never had, audio and video. The integration labor whose forty-five-year residence in human hands Chapter 3 documented is visibly transferred to software, and the export pattern (PPTX, PDF, Sheets at the boundary; grounded corpus inside) is the two-tier format regime of Section 7.4 operating in production: generation lives in the working tier, and boundary formats are compiled only where humans and institutions require them.

Third, Gemini Notebook gives Google a two-surface strategy that Microsoft cannot symmetrically copy. Microsoft's agentic bet runs through one surface: Copilot and Cowork operating inside and alongside the installed apps, preserving the app franchise while changing the worker within it. Google now fields two: Workspace, the artifact-first estate it defends and opens per Section 5.3, and Gemini Notebook, a source-first greenfield unburdened by triad legacy — a surface that quietly renders Google's own editors secondary. Google can afford this internal insurgency precisely because it is the challenger: in the enterprise, it has comparatively little editor franchise to cannibalize. Microsoft could build the equivalent tomorrow; it cannot *lead* with it without detonating the franchise economics its entire defense is constructed to protect. Gemini Notebook is thus the institutional expression of the challenger's freedom — the one position in the field from which the post-app office can be prototyped at scale without self-harm.

The limits are equally instructive, because they mark where the prototype stops short of the endgame. Notebooks are silos: curation is manual and per-notebook, there is as yet no organizational graph knitting notebooks into an estate-wide network, and generated artifacts allow limited post-generation editing — the surface renders and regenerates rather than edits. The bridge from personal notebooks to the organizational knowledge network runs through exactly the assets analyzed elsewhere in this report: Workspace Intelligence for ambient context, Knowledge Catalog for governed enterprise knowledge, and the open format examined in Section 7.6 for the container. Whether Google connects these pieces is among the highest-leverage product questions of the next two years, and Section 8.5 treats one vision of the connected end-state.

5.5 Fortress and Feeding Ground: The Estates as Defended Territory and Raw Material

The arena's final property is its dual character. Each estate is simultaneously a fortress — the territory its owner defends with bundles, gates, and governance — and a feeding ground: the substrate that every third-party agent needs to access in order to be useful, because that is where the users' work and context live (Section 4.2–4.3).

This duality generates the signature relationships of the era, developed in Chapter 6: Anthropic operating *inside* Microsoft's estate both as an invader (Claude add-ins in Excel/PowerPoint/Word/Outlook) and as a supplier (Claude as the engine of Microsoft's own Copilot Cowork); Google laying public roads into its estate for rival agents while racing to keep its first-party agent (Spark) the most natively advantaged driver on them; and every independent harness company's viability depending on gate policies it does not control. The estates are not merely competitors in the agentic war; they are its terrain — and the terrain's owners are also two of its armies. Gemini Notebook adds a final wrinkle to the dual character: it is the one major surface that treats the estates themselves as feeding ground — a notebook ingests Docs, Slides, PDFs, and spreadsheets indifferently, whoever authored them — which is exactly what qualifies it as the prototype of the post-estate office examined in Chapter 8.

6. The Players

With the arena defined, this chapter analyzes the armies. The essential break from the Copilot interlude is shared by all of them: the move from "ask and answer" to "delegate and verify." What differs is position — and therefore strategy.

6.1 Microsoft — Full-Stack Defense

Microsoft's response is the most layered in the field, formalized in **Copilot Wave 3** (announced March 9, 2026), which the company frames as the shift from prompt-based assistance to agentic, multi-step, long-running execution built on Work IQ and multi-model reasoning. Four components matter.

Agent Mode: porting the app franchise into the agentic era. Inside Word, Excel, and PowerPoint, Copilot now persists in the document, workbook, or deck across multiple turns — creating, editing, restructuring, styling, and pulling context from the user's work via Work IQ — a pattern Microsoft has marketed as "vibe working," the office analogue of vibe coding, with frontier reasoning models decomposing complex spreadsheet tasks step by step. Agent Mode is the defensive translation of the installed app base: the apps remain the venue, but the worker inside them changes.

Work IQ: the context monopoly play. Analyzed in Section 5.2; strategically, it is the conversion of the Stage-2 file-format moat into an agent-era context moat — the same network effect, one level up the stack.

The agent economy infrastructure. Distribution (the Agent Store, reachable across Teams, Outlook, Word, Excel, and PowerPoint, with admin review and a registry), creation (Copilot Studio's low-code agent building, with computer-using agents reaching general availability in mid-2026), and governance (Agent 365, "one control center to deploy, organize, and govern every agent — no matter how they were built," sold as a $15/user/month add-on). The pricing architecture is itself a strategy: the new **Microsoft 365 E7** suite — the first new enterprise license tier in roughly a decade, generally available May 1, 2026 at $99/user/month — bundles Copilot, Agent 365, and the security stack, while agents touching tenant data are billed on metered consumption. Seat and consumption models now coexist inside one franchise: the bridge pricing of the transition this report projects in Chapter 8.

The multi-model pivot. The most strategically loaded move. Wave 3 brought Anthropic's Claude into mainline Copilot experiences alongside OpenAI's latest models, and — decisively — Microsoft built **Copilot Cowork** on the technology behind Anthropic's Claude Cowork — rolled out first through Microsoft's Frontier early-access program, its pre-GA channel for Copilot features (a distinct product from OpenAI's identically named enterprise platform, Section 6.3.2) — positioning multi-model choice ("the right model for the job regardless of who built it") as Copilot's differentiation. The substrate of the partnership is a $30 billion Azure compute agreement announced in November 2025, with Anthropic operating as a Microsoft subprocessor since January 2026. Even Microsoft's Researcher agent now splits generation from evaluation across models from different labs. The OpenAI-exclusive era of Copilot is over; Microsoft has converted model supply into a hedged portfolio — and placed its sharpest insurgent inside its own product.

6.2 Google — Full-Stack Offense

Google's response, concentrated in April–May 2026, assembles a vertical stack from silicon to end-user agent: TPUs (infrastructure), the **Gemini Enterprise Agent Platform** (the April 2026 evolution of Vertex AI — a comprehensive platform to build, scale, govern, and optimize agents, including hardened sandboxed workspaces where agents safely run bash commands and manage files), **Workspace Intelligence** (the context layer, Section 5.2), and **Gemini Spark** (the end-user agent, unveiled at I/O in May 2026).

Spark is the fullest expression of the delegate-and-verify paradigm shipped by any of the three majors to consumers and small businesses: a 24/7 personal agent built on Gemini 3.5 and the Antigravity agent harness, running on dedicated Google Cloud VMs so tasks continue with the laptop closed, reachable through a dedicated Gmail address the way one would message a colleague, able to browse through Chrome, organized around Tasks, Schedules, and Skills, integrated out-of-the-box with Gmail, Docs, and the rest of Workspace (the structural advantage rivals must replicate through third-party integrations), extensible over MCP, and governed by explicit user approval for high-risk actions such as sending email. The enterprise variant operates inside Gemini Enterprise across Workspace, custom connectors, and the open web.

Spark is, however, only half of Google's end-user offensive. The other half is Gemini Notebook (Section 5.4), which attacks from the opposite direction: not an agent operating the artifact estate, but a surface that makes the artifact estate optional — and whose bottom-up adoption base gives Google a second, independently growing beachhead in the office layer.

Combined with the open-access gambit of Section 5.3 — the Apache-2.0 CLI, the public MCP server, explicit hospitality to rival agents — Google's strategy reads coherently as the challenger's playbook: own the architecture the future converges on (it already does, per Section 5.1), commoditize the access layer to attract agent traffic into the estate, and win the first-party agent race on native-context advantage. Its historical weakness is unchanged: converting architectural superiority into large-enterprise share against Microsoft's distribution, governance story, and bundling power.

6.3 The Model Labs as Insurgent-Suppliers

6.3.1 Anthropic: Harness-First Entry — and the Competitor-Supplier Duality

Anthropic's path inverts both incumbents': owning no apps, it entered through the agent harness and is invading the office layer from below. Three moves define the strategy, and a fourth defines its strangeness.

Cowork: the generalization of Claude Code. Claude Code (launched as a research preview in February 2025) was built for developers, but users stretched it into general work — organizing files, assembling documents — and in January 2026 Anthropic met them with **Claude Cowork**: the same agentic foundations stripped of the terminal, granted access to a folder, instructed in plain language, executing multi-step tasks (a pile of receipt screenshots into an expense spreadsheet; scattered notes into a report draft) with plan-explanation and check-ins, in an interaction the company describes as leaving messages for a coworker rather than chatting with a bot. Launched as a research preview for top-tier subscribers, it reached general availability on macOS and Windows with enterprise controls (group roles, SCIM, admin analytics) by April 2026, plus a plugin ecosystem in which — tellingly for the verticalization of agents — legal professionals became the most engaged knowledge-work cohort. Commentators noted at launch that this positioned Anthropic in direct competition with Microsoft Copilot for enterprise productivity, and predicted (correctly) that the other labs would follow into the category.

Office add-ins: invading the incumbent's apps. Claude entered Excel in beta (October 2025), opened to Pro-tier subscribers in January 2026, expanded to PowerPoint (February 2026), and on May 7, 2026 reached general availability across Excel, PowerPoint, and Word — with Claude for Outlook entering public beta at the same time, and a single conversation following the user across apps, full shared context (analyze in Excel, build the deck in PowerPoint, synthesize in Word without re-explaining), reusable Skills, template- and formula-integrity preservation as first principles, and MCP connectors to S&P Global, LSEG, Daloopa, PitchBook, Moody's, and FactSet squarely targeting financial and research verticals. The strategic geometry is notable: Anthropic does not ask users to leave Microsoft's estate; it moves in.

MCP: owning the protocol layer. The Model Context Protocol, created by Anthropic and now stewarded by the Linux Foundation, has been adopted as the third-party connection standard by Microsoft's Agent Store and by Google's Spark alike. Holding the de facto interconnect of the agent economy may be Anthropic's most undervalued asset (Chapter 7 returns to why protocol, not format, is where standardization now creates power).

The duality. Anthropic is simultaneously Microsoft's invader (add-ins inside Office, Cowork against Copilot) and Microsoft's supplier (Claude as the engine of Copilot Cowork; Claude selectable in mainline Copilot chat; the $30B Azure agreement; subprocessor status). No prior challenger in this industry's history has occupied both positions at once. The configuration is stable only as long as each side's dependence is mutual — Microsoft hedging model supply, Anthropic buying enterprise distribution — and it is the single relationship most worth monitoring in the entire field.

6.3.2 OpenAI: Workspace Agents, Frontier, and the Fast-Follow Pattern

OpenAI's sequence in this category has run consistently behind Anthropic's — Codex after Claude Code; and after Cowork (January 2026), the February 2026 launch of **Frontier**, an enterprise platform for managing "AI coworkers" with shared business context, execution environments, evaluation, and permissions, followed in April by **Workspace Agents** in ChatGPT: the successor to custom GPTs (which will be deprecated for organizations), agents that take on work across Slack, Google Drive, Microsoft apps, Salesforce, Notion, and other tools, built on Codex underneath, free until May 6, 2026 and credit-priced thereafter. The scale, however, forbids dismissal: enterprise already exceeds forty percent of OpenAI's revenue and is tracked toward parity with consumer by the end of 2026, with the Atlas browser as an additional agent surface and forward-deployed engineering as the enterprise wedge. OpenAI's structural difference from Anthropic is the absence of the supplier role inside Microsoft's product strategy that it once monopolized — the multi-model pivot of Section 6.1 diluted precisely that position.

6.4 The AI-Natives: Genspark, Manus, Skywork — Harness Valuations, M&A, and Geopolitics

The AI-native entrants share a single architecture thesis — value lives in orchestration above interchangeable models — and 2026 priced that thesis.

Genspark built an all-in-one AI workspace on a "Mixture of Agents" architecture that blends ChatGPT, Claude, and Gemini for cross-verification, with a Super Agent that decomposes goals, selects tools, and executes end-to-end: cited research pages, AI Slides (structured decks exportable to PPTX), AI Sheets (a junior data analyst that scrapes, structures, and writes Python in notebooks), and from early 2026 a secure cloud environment ("Claw") in which the agent operates real software interfaces. Founded in 2024 by ex-Microsoft search leadership, it reached unicorn valuation on organic growth — an existence proof that orchestration alone, atop rented models, builds a venture-scale office product.

Manus is the era's cleanest datapoint. Launched March 2025 by Butterfly Effect (Chinese-origin, relocated to Singapore), it built a general-purpose execution agent — datasets into presentations, websites, multi-step workflows, with a Chrome extension for web actuation — explicitly on third-party models (Anthropic, Alibaba among them), differentiating on orchestration, reliability, and speed (its 1.5 release cut average task completion from roughly fifteen minutes to under four). By December 2025 it reported roughly $100M ARR, over 147 trillion tokens processed and 80 million virtual machines created; Meta then agreed to acquire it for a reported $2–3 billion — roughly 20–30x ARR, nine months after launch, for a company with no proprietary frontier model. The market had priced the harness above the model. Then geopolitics intervened: China's Ministry of Commerce opened an export-control review in January 2026 and on April 27, 2026 ordered the acquisition unwound — establishing that agent technology has joined semiconductors in the category of geopolitically controlled technology, a risk now attached to every cross-border agent asset.

Skywork (Kunlun Tech) is the Chinese-origin "AI version of Office" played globally: Super Agents launched May 2025 with five expert agents — Documents, Slides, Sheets, Web, Podcast — plus a general agent, a DeepResearch engine cross-validating hundreds of sources per task with traceable citations, top GAIA benchmark claims, a mobile app marketed as "eight hours of work in eight minutes," and from February 2026 a Windows desktop application positioned squarely as a Cowork alternative — targeting the world's largest productivity platform while integrating both Google's Gemini and Anthropic's Claude as its model layer. Like Manus, like Genspark: a harness company, models rented.

The collective lesson of 6.4 reinforces 6.3: in the agent era a growing share of differentiated engineering value resides in the harness, and incumbency in models is no protection against — indeed is a supply input to — harness competitors.

6.5 Adjacent SaaS Reinventions: Notion and Canva

The Stage-3 survivors are not waiting to be disrupted.

Notion executed the most aggressive agent conversion of any incumbent SaaS. Version 3.0 (September 2025) rebuilt the product around autonomous agents capable of twenty-plus minutes of multi-step work across hundreds of pages; 3.3 (February 2026) shipped **Custom Agents** — autonomous, schedule- and trigger-driven teammates running 24/7, shareable across teams, permissioned like employees, connected over MCP to Slack, Figma, Linear, HubSpot, Salesforce — with prompt-injection guardrails and full run logging. Two datapoints stand out. Adoption: testers built over 21,000 agents within weeks, and Notion itself runs roughly 2,800 internal agents — more agents than it has employees, the emblem of the "more agents than humans" workplace. Pricing: seats stay priced as before, while Custom Agents consume **Notion Credits** from May 4, 2026 (roughly $10 per 1,000 credits, metered per run) — the cleanest live experiment in the seat-plus-consumption hybrid this report projects as the transition-era revenue model.

Canva, the Stage-3 niche survivor that escaped the bundle by creating a category, is now expanding the category into an operating system. The **Creative Operating System** (October 2025) unified its Visual Suite, AI layer, and platform for 260M+ users; made the professional Affinity suite free forever (a direct strike at Adobe); and introduced a proprietary Design Foundation Model generating *editable* designs plus an on-demand design assistant. In 2026 it acquired Cavalry and Mango AI (February), then Simtheory (agent management) and Ortto (customer data and marketing automation) in April, extending from design into the full marketing lifecycle, and shipped an agentic "AI 2.0" platform with prompt-based editing and persistent memory. Canva illustrates the survivor criterion of the fourth bundling war: do not defend a feature; own an adjacent operating system the bundle cannot cheaply replicate.

6.6 The Interaction Map: Dualities, the Fourth Bundling War, and Agent Traffic Flows

Three interaction patterns organize the field.

Supplier–competitor dualities. The era's signature relationship structure. Anthropic powers Copilot Cowork while invading Office (6.3.1). Manus, Skywork, and Genspark run on models from labs that compete with them downstream. Microsoft funds, hosts (Azure), and embeds the very lab whose Cowork defines its competitive frontier. Google paves roads for rival agents into the estate its own agent must win. In a stack this interdependent, classical exclusive-alliance logic (the OpenAI–Microsoft model of 2023) has given way to portfolio logic on all sides — every player hedges, and the hedges interlock.

The fourth bundling war. Microsoft has now run the bundling play four times: Office against the standalone champions (Stage 2), Teams against Slack (Stage 3), Copilot against point AI tools (the interlude), and E7/Agent 365 against the agent-era field (now). The first three succeeded. The fourth differs in two respects that keep its outcome genuinely open: the components being bundled include *rivals' engines* (Claude inside Copilot) whose makers retain independent channels to the same end users; and the bundle's targets — the model labs — possess what prior bundling victims never had, namely direct consumer relationships of enormous scale and a technology layer (the harness) the bundler must buy rather than build. Regulatory attention to AI-era bundling, already primed by the Teams precedent in Europe, is the additional unknown.

Agent traffic flows across the estates. The forward-looking variable. As work delegated to agents grows, the question "which estate do agents find easier and richer to operate?" becomes a market-share force in its own right (Section 5.3). Google has optimized for agent-traffic acquisition (open CLI, public MCP, third-party hospitality); Microsoft for agent-traffic monetization (credits, governance, context premium). If third-party agent traffic visibly concentrates on the easier estate, Microsoft faces pressure to lower its gates; if enterprise buyers prove to prefer governed, metered paths, Microsoft's model becomes the standard and Google's openness a subsidy. Disclosures of credit consumption and agent-traffic mix in late 2026 will be the first hard evidence either way.

7. Files, Formats, and Protocols

The file format was this industry's original weapon. This chapter traces its disarmament, identifies the layers to which its power has moved — and closes with the construct toward which the format story now points: the enterprise knowledge network, and the new class of asset it creates.

7.1 A Brief History of Format Power: Binary Moat → OOXML Opening → File-less Cloud

In Stage 2, .doc and .xls were proprietary binary formats whose network effect — documents must be exchangeable, therefore everyone needs the software that reads what everyone sends — constituted the core of the Office moat. The first demotion came in the mid-2000s: under pressure from the OpenDocument camp and government-procurement regulation, Microsoft opened its formats as OOXML, standardized as ECMA-376 (2006) and ISO/IEC 29500 (2008) after one of the most contentious standardization fights on record. The format moved from moat to compatibility layer. The second demotion came with the cloud: in Google's architecture (Section 5.1) the file disappeared entirely — a database record with export options. The strategic status of the file format, in other words, was already dying before agents arrived; the agentic era completes the execution.

7.2 OOXML as a Neutralized Output Codec — and the Residual Gravity of the Reference Renderer

What exactly is OOXML's openness worth? The .docx/.xlsx/.pptx containers are ZIP archives of XML, and anyone may read and write them royalty-free — LibreOffice, Google's import/export, Apple iWork, and the open-source libraries (python-docx, openpyxl, python-pptx) through which today's agents generate Office files without owning Office. But three asymmetries qualify the neutrality. First, the standard split into Strict and Transitional variants, and Microsoft Office writes Transitional by default — a variant carrying legacy elements that effectively reference decades of Word's and Excel's historical behavior, so that full fidelity requires reverse-engineering Microsoft's past. Second, and decisively, the specification defines *editable structure*, not *rendering*: line-breaking, pagination, and font metrics are left to the application, which is why the same .docx can paginate differently in Word and LibreOffice. The contrast is PDF (ISO 32000, control relinquished by Adobe), which specifies the final rendered result and therefore displays virtually identically everywhere. OOXML's de facto reference implementation is not the standard document; it is Microsoft Office — "does it open correctly in Word" remains the practical definition of compatibility. Third, consequential territory (VBA macro behavior, some embedded objects) sits outside the standard's full specification, and the specification's sheer mass (thousands of pages) is itself an implementation barrier.

The agent-era synthesis follows directly. Because generation is commoditized, *adopting OOXML as an output format* carries almost no lock-in — the PDF analogy holds: everyone outputs it, no one is captured by it. Every player accordingly converges on OOXML as the **neutralized output codec** for the human-institutional boundary, while keeping their *working* representations elsewhere. Microsoft's residual gravity is precisely and only the reference-renderer position — which lasts exactly as long as human verification happens on a Word or Excel screen, and decays as that verification migrates to agent interfaces (Chapter 8).

7.3 The Decomposition Thesis: Data + Code + Web Rendering vs. the Fused Container

The .xlsx and .pptx containers are not mere files: the spreadsheet fuses data with logic (formulas, dependencies, pivots, macros — a spreadsheet is a program), and the presentation fuses content with behavior (animation, transitions). Even with OOXML open, the semantics of that fused logic are best understood by its author, leaving Microsoft a residual pull inside the container.

The AI-native escape route is decomposition: data in open formats (CSV at the low end, Parquet/Arrow at scale, JSON for structure), logic as universal code (Python, SQL), rendering on web standards (HTML/CSS/JS) — the open web substrate, which stands as the third architecture alongside the app-native container and the file-less service of Section 5.1. Each decomposed component is an unowned substrate — Python formulas need no Microsoft calculation engine; CSS animation needs no PowerPoint XML — and, critically, code is the representation LLMs handle best: testable, diffable, version-controlled, self-verifiable by the agent. The decomposition is not hypothetical; it is the observed behavior of the AI-native products (Genspark's Sheets agent writes Python in notebooks rather than spreadsheet formulas; the AI-native presentation tools render decks as web pages, exporting to .pptx only at the boundary). The inversion can be stated in one line: from *the document that computes* to *the program that renders the document*.

Three counterforces complicate the picture. (1) **The verification asymmetry:** Excel's formula language is the only programming language the world's non-programmers can read. Decomposing a financial model into Python delights the agent and burdens the CFO, the auditor, and the regulator; in precision-and-accountability domains the fused grid persists as institutional infrastructure, relaxing only as fast as verification itself passes from humans to agents — which makes the verification surface, like the runtime, a target to which lock-in relocates (Section 8.5 develops it into the audit chair). (2) **The runtime dependency:** the fused container is self-contained — open it and it computes — whereas data-plus-code requires an execution sandbox. Lock-in does not vanish; it migrates from the container to the runtime, which is exactly why sandboxed agent workspaces (Google's Agent Platform, the labs' execution environments) are being productized as strategic ground. The old question "whose format stores the work?" becomes "whose sandbox executes it?" (3) **Microsoft's re-absorption counter:** Python in Excel (2023–) and Office Scripts pull the code layer *inside* the grid — the decomposition's expressive power wrapped in the fused model's verification shell. Whether decomposition or re-absorption wins is governed by counterforce (1): the speed at which the verifier stops being human.

Spreadsheet-format futures therefore resolve not to a single successor but to a division of labor: .xlsx as the human audit surface, Parquet for data interchange, SQL/code for analytical logic. CSV remains ubiquitous for simple interchange, but its lack of explicit types, schema, and embedded logic makes it increasingly inadequate for agent-mediated analytical work — squeezed out of the working tier by Parquet at scale.

7.4 Markdown as the Agent Lingua Franca: The Two-Tier Format Regime

Will Markdown become the new document standard? Half-yes — and the half matters. Markdown's properties (an order-of-magnitude token efficiency over HTML, deterministic parsing, diff-ability, human readability, and saturation of LLM training distributions) have made it the de facto *working language* of agents along two visible tracks. As the **instruction and context format**: AGENTS.md has spread across tens of thousands of repositories and entered the Linux Foundation's agentic-AI umbrella, while CLAUDE.md and SKILL.md files (the latter adopted even by Google's Workspace CLI for its bundled skills) have made Markdown effectively the programming language of agent configuration. As the **machine-facing twin of the web**: the llms.txt convention (proposed September 2024) reached roughly ten percent of major domains by early 2026, Cloudflare ships automatic Markdown conversion for agent crawlers at the CDN layer, and agent crawlers preferentially ingest Markdown when sites serve both — the web bifurcating into a human surface (HTML) and a machine surface (MD).

What Markdown cannot do is replace the boundary formats. It lacks fixed layout, pagination, tracked changes, signatures, and legal rendering guarantees (the *record* half of the document axis stays with PDF/DOCX); it has no real presentation equivalent; and CSV, its tabular cousin, falls short per 7.3. The stable structure is therefore a **two-tier format regime**: an agent-native working tier (Markdown for narrative and instructions, JSON for structure, Parquet/SQL for quantitative work) in which agents think and exchange, and a human-institutional boundary tier (DOCX/PDF for record, XLSX for audit, PPTX for persuasion) into which work is *compiled* wherever humans and institutions require it. The source-versus-binary analogy is exact: Markdown is the source; OOXML is the build artifact. As Phase B (Chapter 8) shrinks the human boundary, the boundary tier's generation frequency shrinks with it.

The strategic punchline: Markdown wins *because* it is unownable — and for the same reason its victory generates no moat for anyone. Format standardization is occurring, and it is strategically sterile. (Section 7.6 qualifies this verdict in one respect: the format is sterile, but the *conventions built on it* — and above all the governed serving of them — are not.)

7.5 From Format Wars to Protocol Wars

If formats no longer carry power, where did the power go? One layer up. Agent-to-agent and agent-to-estate exchange happens through calls, not files, so the 1990s .doc war's true successor is the contest over **interfaces and schemas**: Work IQ's API as the gated, monetized door to Microsoft's context; Google's API-as-format architecture, now with public MCP on-ramps; MCP itself — created by Anthropic, stewarded by the Linux Foundation, adopted across rival ecosystems — as the open interconnect; A2A for agent delegation; and, in the commerce theater where the stakes are most visible, an open protocol contest among Google's Universal Commerce Protocol, OpenAI and Stripe's Agentic Commerce Protocol, and the Google-led AP2 payment-authorization layer — developed with a broad coalition of payment networks and technology companies — with MCP and A2A as adjacent interoperability layers underneath. Add the two relocation targets identified in 7.3 — execution sandboxes and verification interfaces — and the full map of where lock-in now lives is drawn: **protocols, context graphs, runtimes, and the surfaces where humans check the machines.** The owner of OOXML watches its format moat evaporate; the owners of protocols and context graphs watch theirs fill. (The protocol theater is the subject of this series' Report 5, *The Agentic Mesh*; the remainder of this chapter pursues the format story to its own terminus.)

7.6 The Agent-Native Knowledge Stack: OKF, Parquet, and the Unbundling of the Formula

On June 12, 2026, the working tier of Section 7.4 acquired a formal specification. Google published the **Open Knowledge Format (OKF)** — an open, vendor-neutral v0.1 spec representing organizational knowledge as a directory of Markdown files with YAML frontmatter: one concept per file, the file path as the concept's identity, ordinary Markdown links turning the directory into a knowledge graph, index files for progressive disclosure, logs for change history. No SDK, no runtime, no registry; readable in any editor, shippable in any Git repository. The spec explicitly formalizes the "LLM wiki" pattern — Andrej Karpathy's observation that the maintenance drudgery which defeats human wiki-keepers is exactly what LLMs perform tirelessly — that practitioners had been reinventing ad hoc through AGENTS.md files, Obsidian vaults, and metadata-as-code repositories. Google's Knowledge Catalog both produces and ingests OKF bundles, and ships a reference agent that walks a BigQuery dataset drafting a concept document for every table. This series' Report 6, *The Data Layer Reforged*, analyzes OKF from the data platform's side, under the name *compiled context*; this section analyzes what it means for the office layer.

Read against the triad, OKF completes a picture this chapter has been assembling piecewise. Each axis of the triad now has a named agent-native counterpart. The **document axis** maps to OKF Markdown: narrative knowledge as linked, typed, versioned concepts. The **spreadsheet axis** maps to Parquet plus code, per the decomposition of 7.3 — with a token-economic logic worth stating plainly. An agent that reads a hundred-thousand-row workbook as text pays for every cell in context-window tokens, a cost that scales linearly toward context exhaustion; an agent that writes a query against a columnar Parquet file pays context-window costs only for the code and the summarized result — a cost largely independent of the dataset's total size — while execution cost, which still scales with the data scanned, is held down by the format's compression, predicate pushdown, and column pruning. Practitioner literature circulates large headline figures for the token savings of Markdown over markup and code-over-cells over cell-reading; the specific numbers are unaudited and should be treated as directional, but the direction is not in doubt, because it follows from the arithmetic of tokenization itself. The **presentation axis** maps to JSON state plus web rendering, per 7.3. The working tier, in other words, is no longer a loose habit; it is a stack — MD/Parquet/JSON — with an interoperability spec at its qualitative corner.

The genuinely novel element this stack enables is not the container but a pattern best called the **metric concept**: business logic itself — the definition of a KPI, the formula for a churn rate, the rule for revenue recognition — expressed as a governed Markdown concept. OKF makes the pattern possible without prescribing it: in the v0.1 specification the only required frontmatter field is `type`, with `resource` available as an optional pointer to the governed dataset — so a producer can represent a KPI as a `type: Metric` concept, bind it to its Parquet resource through `resource`, add producer-defined fields such as `owner` and `status`, and carry the formal mathematical definition in the Markdown body. Readers of Report 6 will recognize the object: it is the semantic layer's metric view — Unity Catalog Business Semantics, Snowflake's semantic views — externalized from the platform into portable text. And it bears directly on the verification asymmetry that Section 7.3 identified as the decomposition thesis's chief counterforce. The Excel defense holds that the grid is auditable by non-programmers while Python is a black box; the honest rebuttal is that real corporate spreadsheets decay into nested formula sprawl across hidden tabs, that the grid fuses data and logic in the same cell so a hardcoded override breaks the audit trail silently, and that the grid enforces no types. The metric-concept pattern resolves the standoff by unbundling the Excel cell into three separately governable layers: **data** (immutable, typed, columnar — Parquet), **logic** (declared, versioned, human-readable — the OKF concept), and **execution** (generated, disposable — code the agent compiles from the declaration, runs against the data, and explains back). The human audits the declaration; the agent manufactures and justifies the computation; neither audits the other's weakest surface. Note the direction of this argument: it does not merely blunt the Excel defense but inverts it. If audit migrates from the grid to a rendered view of a declared concept, then verification — the spreadsheet's last irreplaceable function (Section 7.3) — has left the spreadsheet. Whatever the new audit surface turns out to be, the grid's institutional privilege dissolves the moment checking no longer happens inside it; the strongest remaining case for the fused container is thereby weakened, not reinforced.

One caution belongs here. OKF has shipped in its earliest and simplest form — a v0.1 specification whose adoption beyond Google's own tooling is not yet evidenced — and formats are decided by defaults, not by design merit: its fate turns on whether the tools that generate enterprise knowledge, from Cowork and Copilot to Gemini Notebook and Notion, come to emit and consume it by default. That verdict is years, not quarters, away, and the pattern described here should be read as a credible trajectory to monitor rather than a settled outcome. A second caution is architectural rather than adoptive: a compiled knowledge estate is a second copy of the organization's truth, and the moment it drifts from its sources the enterprise operates on two — the **dual-truth problem**, the standing synchronization cost of any compiled-context architecture (Report 6 reaches the same conclusion from the data platform's side), whose operational and risk consequences Sections 8.5 and 9.4 carry forward.

7.7 Knowledge Capital: What the Network Creates

If the stack of 7.6 is adopted in any of its forms, the office layer's output changes kind. For forty-five years, knowledge work produced *artifacts* — documents, workbooks, decks — that encoded knowledge in forms only humans could reassemble. A knowledge network produces something categorically different: the organization's knowledge as a **curated, linked, machine-executable estate** — qualitative concepts cross-referenced into a graph, quantitative data typed and columnar beneath them, business logic declared between the two — that agents can traverse, compute against, and act from. This report names the asset that estate constitutes **knowledge capital**, and offers three propositions about it.

It is the refined form of an asset the LLM era just revalued. Report 6 argued that language models made a far larger share of the enterprise's qualitative corpus — the majority of corporate information, stored as language — economically and interactively computable through a general-purpose interface, among the largest revaluations of a dormant corporate asset class in the history of enterprise IT. Raw computability, however, is crude oil; the knowledge network is the refinery. Two firms with identical headcounts and identical model subscriptions will diverge on the quality of the knowledge estate their agents work from — and that divergence compounds, because an agent-maintained network improves with use: every task that traverses it can enrich it, every cross-reference an agent repairs stays repaired. Agent maintenance can drive the marginal cost of keeping the network current far below what human curation ever achieved — validation, governance, and synchronization remain durable costs — and that economics distinguishes knowledge capital both from raw data (unrefined) and from human capital (which resigns).

Its formation will follow one of three paths. The paths differ on a single question: where the organization's knowledge ends up living, and whether it can be carried elsewhere.

K1 — the estate-compiled network (in this report's judgment, the dominant path). The company moves nothing. Its knowledge stays where it already sits, and the incumbent's tools — Work IQ, Workspace Intelligence, Knowledge Catalog — read the existing pile of documents and data in place and turn it into an organized, governed knowledge graph. Open bundles (OKF and its kin) are used only for export and exchange at the edges, not as the place the real knowledge lives. The appeal is that enterprises get a knowledge network without a migration. The cost is that lock-in does not disappear; it changes form. Where a company was once held by its file formats, it is now held by the knowledge map its vendor built for it — the *semantic lock-in* that Report 6 identifies as the successor to data lock-in. This is the path with the most evidence behind it, because it is already shipping: Knowledge Catalog's ingestion of OKF, and the whole design of Work IQ, are this scenario in production.

K2 — the portable network (a meaningful and lasting minority). The company deliberately keeps its knowledge vendor-neutral — plain Markdown for concepts and Parquet for data, versioned in Git — so that it can be moved to any tool at any time. Editors such as Word and Google Docs are demoted to interchangeable windows onto that knowledge; the knowledge itself is portable by construction. The constituencies that want this are real: developer-led firms already living in Git, regulated and sovereign buyers who must be able to walk away with their records intact, and AI-native companies built on the stack from birth. But keeping such an estate curated and governed is real work, so in practice most who choose this path will run it *through* a commercial governed layer rather than entirely on their own.

K3 — fragmented drift (the residual, unhappy case). The open format never catches on — OKF joins OpenDocument in the museum of well-designed standards that lost to defaults — and every vendor's knowledge stays trapped in its own system, unable to travel. Companies accumulate knowledge capital everywhere and can move it nowhere, and the cost of switching vendors climbs to its historical maximum.

The realistic end-state is a blend of the first two, in a shape we have seen before. Rather than one path winning outright, the likely outcome is **K1 at the core and K2 at the edge**: the vendor's proprietary knowledge graph sits at the center of the enterprise, while open bundles handle exchange at the boundary — a governed core wrapped in open interchange. The reason to expect this specific equilibrium is that the data-infrastructure world already ran the experiment and landed exactly here (Report 6, Chapter 10): the open table format made raw *storage* neutral and portable, yet the governed catalog sitting above it became the real point of control and the new source of lock-in. The office layer is on track to reproduce that outcome with the parts renamed — the knowledge bundle playing the role the table format played, and the context graph playing the role of the catalog. The naive hope that open formats will simply set enterprise knowledge free is therefore only half right: the *format* will open, but the layer that organizes and governs the knowledge above it will remain proprietary ground, and that is where both the lock-in and the value will move. How much weight each path deserves, the signals that would shift those weights, and what all of this means for Microsoft and Google are the business of Section 8.5.

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8. Scenarios and Outlook

The transformation will not arrive as a single discontinuity. Three phases, separated by where the source of truth lives and where the human stands.

8.1 Phase A — The Actuator Era (Now–2027): Agents Operate the Existing Apps

The current state, fully described by Chapters 5–6: agents work *through* the installed triad — Agent Mode inside the Office apps, Claude add-ins operating Excel and PowerPoint, Spark acting across Workspace, harness startups driving the suites through APIs, CLIs, and browsers. The triad apps retain their value but change character: from the human's workspace to the agent's actuator plus the human's review screen. Files remain the persistence layer; OOXML output is everywhere and locks in no one (7.2); per-seat pricing holds, with consumption meters attached at the edges (Copilot Credits, Notion Credits, OpenAI agent credits). Phase A's competitive question is the gate question of Section 5.3 — and its monitoring variable is agent traffic share by estate.

8.2 Phase B — The Rendering Era (Toward 2030): Files Become Views, Seats Become Credits

The structural break. The source of truth migrates from files to the context substrate — the organizational graph plus the agent-native working tier of 7.4 — and triad artifacts become on-demand renderings: compiled views, generated at the moment of human need, increasingly disposable afterward. ("Which version is current?" stops being a question; the graph is current, files are snapshots.) Google's architecture, which embodied this model two decades early, becomes the pattern everyone — Microsoft via Work IQ most deliberately — converges on. Chapter 7 gave the substrate its concrete form: in practice, the context substrate *is* the knowledge network of Section 7.7 — governed context in the estates' compiled graphs, interchanged at the boundary as agent-native bundles (OKF-class Markdown for qualitative knowledge, Parquet for quantitative data, JSON state for the visual layer) — and Gemini Notebook (Section 5.4) is what the human-facing surface of such a substrate looks like when shipped as a product.

Phase B is where per-seat economics break in earnest, because the model's foundation — counting humans who use apps — decouples from where work happens. The transition pricing is already legible in 2026's experiments: the hybrid of retained seats plus metered agent consumption (Notion's seats-plus-credits, Microsoft's E7-plus-metered-agents, OpenAI's credit-priced Workspace Agents). The hybrid is the bridge; the consumption share of revenue is the bridge's progress meter. Phase B is also where the two-tier format regime matures: boundary-tier artifacts are generated less frequently, working-tier interchange dominates, and the reference-renderer gravity of 7.2 visibly decays as verification moves into agent-mediated interfaces.

8.3 Phase C — The Human-Boundary Contraction: Office TAM Redefined by Attention

The structural endpoint. Agent-to-agent collaboration needs no triad at all — machines do not persuade each other with slides — so triad artifacts come to be generated *only where a human stands in the loop*: to inform a decision, to satisfy an audit, to persuade a board, to constitute a record. The triad's addressable market is thereby redefined: not the volume of knowledge work (which agents inflate) but the area of the human-attention boundary (which delegation contracts). The office category does not disappear — Section 4.5's function-persistence argument guarantees the functions' survival — but its product form is fully inverted: from the universal workspace of humans to the interface layer between an organization's people and its machine workforce.

8.4 Differential Fates of the Three Axes

The axes do not decay uniformly, because their human functions differ.

Presentation: fastest production commoditization, persistent function. As the pure rendering of reasoning, deck production automates first (the Gamma/Canva/AI-slides explosion is the leading indicator). But its real function — human-to-human persuasion — survives as long as humans remain the audience for decisions. Production labor value goes to zero; what remains monetizable is what to argue, plus adjacent assets (brand governance, design systems) of the Canva pattern.

Spreadsheet: the most durable axis. Paradoxically strengthened: the grid becomes the audit surface for agent work — the near-unique interface on which non-programmer humans can inspect machine quantitative reasoning (the verification asymmetry of 7.3, and the reason Anthropic leads its Excel pitch with formula-and-dependency preservation and financial-data connectors). Authorship passes to agents; the format's institutional standing — finance, audit, regulation — outlasts both siblings for as long as verification remains a human act performed on the grid. That condition is precisely what the metric-concept argument of Section 7.6 attacks and what indicator 8.6(3) measures: the axis's durability is real but conditional, its horizon set by the pace at which audit migrates to rendered-concept surfaces (Section 8.5).

Document: the axis that splits. Its communication function (memos, status updates, information transfer between humans) dissolves fastest — in a world where agents read and write the context directly, the intermediate document loses its reason to exist. Its record function (contracts, policies, evidence of decision and responsibility) persists and acquires a growth driver: the official record of what the agents did. Document-axis TAM shrinks; record-of-truth value compounds, fused to the context layer.

8.5 The Knowledge-Network Endgame: One Vision of the Duopoly's Adaptation

What follows is a scenario, not a forecast: one internally consistent end-state for how the two estates adapt if the knowledge network of Section 7.7 becomes the office layer's organizing construct. It is presented because a vision-level view of the terminus disciplines the near-term analysis — and its load-bearing assumptions are listed at the end, so the reader can watch them fail.

The scenario's premise. The two-tier regime of 7.4 matures past coexistence: the working tier — the knowledge network — becomes the enterprise's operative source of truth for a widening share of knowledge, while boundary artifacts are compiled on demand. Under that premise, the duopoly's adaptation paths diverge according to what each already owns.

Microsoft: the both-sides incumbent. The naive version of the open-network thesis casts Microsoft as the format transition's loser — the owner of the fused container watching truth migrate to plain files. That version misreads the portfolio. Microsoft owns the open stack's natural habitat: **GitHub**, the world's largest Markdown-and-Git estate and the de facto home of the agent-configuration conventions (AGENTS.md and kin) from which the knowledge-network pattern grew; **VS Code**, the default editor of the working tier; **Windows and WSL**, the obvious host of the local execution sandbox that the decomposition thesis (7.3) identified as the new runtime dependency — the place where Parquet-scale computation runs against local data without a cloud round-trip; and **Azure**, the cloud half of the same sandbox. In this scenario Microsoft's predictable play is *compile and absorb*: Work IQ operates as the compiler of the vast installed OOXML estate into governed context; Office's applications are repositioned — not retired — as the premium rendering and **verification** surfaces of network content; and the local sandbox becomes the new attach point of the Windows franchise. The Microsoft of this scenario is a runtime, governance, and verification company whose office revenue has migrated into E7-class bundles plus agent consumption — a company that sold the replacement for its own franchise, because it happened to own the replacement's parts. Its risk is not capability but *pacing*: the innovator's-dilemma clock, moving at the speed of franchise protection rather than at the speed of the technology. The tell will be whether Office-native knowledge-network features (Markdown-native authoring, Git-backed knowledge bases, bundle import/export in Work IQ) ship as first-class products or as containment features.

Google: the standard-setter's gambit, with Gemini Notebook as the front-end. OKF, read strategically, is an attack on the rival's residual gravity that costs its author almost nothing: standardizing the knowledge container erodes the format-and-renderer gravity that is Microsoft's residual moat (7.2), while conceding nothing Google relies on — Google's lock-in never lived in formats (5.1: the format is the API), and its compiled graph remains proprietary behind the open boundary. In this scenario Google's assembly is: Workspace Intelligence and Knowledge Catalog as the governed graph; Workspace as the collaboration surface; and **Gemini Notebook as the knowledge network's human front-end**, evolving along a three-step path of which the first steps are already shipped. Step one, the source-grounded studio of Section 5.4 — corpus in, rendered views out — is production reality. Step two, projection with near-term plausibility: notebooks *mount* living knowledge estates rather than ingesting static uploads — a Drive folder, a Git repository, an OKF bundle — so the notebook's corpus stays synchronized with the organization's network, and the YAML-and-links structure is traversed as a pre-indexed graph rather than re-derived from raw text. Step three, the speculative link: execution — the notebook that, asked a quantitative question, locates the governing metric concept, compiles its declared definition into code, runs it against the linked Parquet resource in a managed sandbox, and returns the answer with the computation's justification, closing the loop that Section 7.6 called the unbundled formula. Every ingredient exists somewhere in Google's 2026 portfolio (Data Tables flowing to Sheets, PPTX compilation at the boundary, Gemini-queryable notebooks, the Agent Platform's sandboxed workspaces — and, since the

July 2026 rebrand, native code execution inside the notebook itself: the per-notebook “secure cloud computer” that writes and runs code against a notebook’s sources); what the scenario asserts, and the record does not yet, is their assembly into one governed surface — the execution substrate now ships, but the metric-concept linkage that would make it the unbundled formula of Section 7.6 does not. Google’s risk in this scenario is the mirror of Microsoft’s: not pacing but *monetization* — the open-network-plus-free-surface pattern acquires users magnificently and converts them through exactly the enterprise sales motion where Microsoft’s machine is strongest.

What the scenario does to the editor-commoditization claim. The strong form of the open-network thesis holds that when truth lives in plain files, the editor is commoditized into an interchangeable renderer. This report endorses half of it. Editors are indeed demoted as *authoring* surfaces — agents author, and authorship is where the triad’s labor value lived. But the same logic that demotes authoring elevates **verification** (4.4, 7.3): in an agent economy, the surface where the human inspects, challenges, and signs off on machine work is not a commodity — it is the audit chair, the point where governance premiums are collected and where the estates’ institutional trust is hardest to replicate.

That audit chair is likely to cohere around four properties. It is **reasoning-first** — reviewing *why* an agent reached a result (the declared metric concept of 7.6, with data, logic, and execution held separately) rather than the finished artifact. It is **diff- and provenance-driven** — foregrounding what changed and where each claim traces to, since no human reviews machine output in full. It is **exception-based** — an approval queue that surfaces only low-confidence, policy-adjacent, or anomalous items. And it is the seat of **sign-off** — where a human assumes responsibility, and where governance premiums and hard-to-replicate institutional trust accrue. Whether this surface is the spreadsheet grid reborn in verification mode or a new renderer that displaces it is the open question — and the one indicator 8.6(3) is built to settle. “Commoditized to a renderer” understates what a renderer is about to become.

Load-bearing assumptions — the scenario’s falsifiers. (1) OKF-class conventions achieve meaningful multi-vendor adoption within roughly twenty-four months; failure collapses this scenario toward K3 (fragmented drift). (2) Enterprises accept bundle- or repository-resident knowledge for at least their non-regulated domains; failure collapses it toward pure K1 (estate-compiled, closed). (3) Agent-maintained bundle freshness proves reliable enough that the dual-truth problem of 7.6 stays operationally tolerable. (4) Verification genuinely migrates to rendered-concept surfaces rather than remaining on the legacy grid. Each is observable; none is yet established.

8.6 Leading Indicators to Monitor

Eight measurables will date the phase boundaries and arbitrate the scenarios. (1) **Credit-revenue mix:** the consumption share of Microsoft 365/Copilot, Notion, and OpenAI enterprise revenue — the direct meter of the seat-model transition. (2) **Agent traffic share by estate:** third-party agent activity on Google’s open gates versus Microsoft’s metered ones; visible concentration forces the laggard’s gate policy to move. (3) **Verification-interface migration:** evidence (product telemetry, enterprise surveys) that review of agent output is shifting from Office screens to agent-native interfaces — the decay rate of the reference-renderer moat and of the spreadsheet’s audit privilege. (4) **Artifact ephemeralization:** the ratio of agent-generated to human-authored Office artifacts, and their retention half-life — the empirical signature of Phase B. (5) **The Microsoft–Anthropic relationship:** renewal, deepening, or rupture of the competitor-supplier duality; any change re-prices the model labs' enterprise distribution and Microsoft's hedge simultaneously. (6) **OKF adoption beyond Google:** bundle emission or ingestion by any non-Google tool of consequence — Cowork, Copilot, Notion, an independent catalog vendor — is the single sharpest arbiter between the K1, K2, and K3 paths of Section 7.7. (7) **Gemini Notebook's estate integration:** organizational notebook sharing at scale, live mounting of Drive- or repository-resident sources, and enterprise attach rates — the milestones of the front-end trajectory in Section 8.5. (8) **Knowledge-network features inside the suites:** Markdown-native authoring, Git-backed knowledge bases, or bundle import/export shipping as first-class Office or Workspace capabilities — the tell for whether the incumbents are absorbing the pattern or containing it.

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9. Investment Implications

9.1 The Duopoly: Defense Quality Assessment

Microsoft is executing the strongest incumbent defense in the category's history, and the historical base rate favors it: the company has now survived four platform transitions (GUI, cloud, collaboration, generative AI) with a different weapon each time. The current arsenal — Agent Mode translating the app franchise, Work IQ converting the format moat into a context moat, E7/Agent 365 running the fourth bundle, and a hedged multi-model supply chain — is coherent and mutually reinforcing. The honest bull case is that the gated-monetization posture wins if enterprise buyers prove to prefer governed, metered agent paths; the honest risks are (a) that gating cedes the agent-traffic flywheel to Google, (b) that the fourth bundle, unlike the first three, draws decisive regulatory interference, and (c) that the per-seat-to-consumption transition is dilutive in the interim even when executed well — bridge pricing (E7 at $99) is designed to mask exactly this, and the credit-mix disclosure of indicator 8.6(1) is the tell.

Google holds the architectural high ground (the file-less, API-gated model the industry is converging toward) and the most coherent challenger strategy (commoditize access, win on native context, field the most advantaged first-party agent). Its open-access gambit is genuinely asymmetric: every unit of third-party agent traffic it attracts is both a subsidy to rivals' agents and a deepening of its own estate's centrality. The persistent bear case is unchanged from Stage 3 — architectural superiority has never yet converted into large-enterprise share against Microsoft's distribution and governance story — and the openness is already showing meters at the margin (API tiering). Google is the higher-variance position on the same trade.

The knowledge-architecture overlay. The analysis of Chapters 7 and 8 adds a dimension to the duopoly assessment that the app-and-agent framing misses, and it cuts in both directions. For Microsoft, the overlay is a hedge the format-war narrative ignores: through GitHub, VS Code, and the Windows/WSL execution layer, Microsoft owns the natural habitat of the open knowledge stack — the Markdown-and-Git estate, its default editor, and its local sandbox — which means the scenario most threatening to the Office franchise is one whose infrastructure Microsoft would supply. The market prices the threatened franchise; it receives the successor stack at no separately visible charge. For Google, the overlay is optionality: Gemini Notebook is the field's only shipped source-first office surface, acquired bottom-up at consumer scale before the enterprise contest has begun, and OKF is a low-cost option on setting the knowledge container's standard. Neither is reflected in how the office contest is conventionally scored. Reframed at the level this report considers fundamental, the duopoly contest is less "who defends the office apps" than **"who compiles the enterprise's knowledge capital"** — Microsoft compiling from the largest artifact estate under the strongest governance franchise, Google from the cleanest architecture with the best greenfield surface. On that framing both estates are structural winners of the knowledge-network transition against the rest of the field, and the interesting variance between them is pacing risk (Microsoft, per Section 8.5) against monetization risk (Google, per the same).

9.2 Model Labs and Harness Players: Where the Manus Multiple Points

The Manus transaction — roughly 20–30x reported ARR, nine months post-launch, no proprietary model — is the cleanest market price yet printed on the proposition that the harness, not the model, is the scarce engineered asset of the agent era. Three implications. First, model capability is being commoditized *from above* by the labs' own competition, while harness quality (reliability on long tasks, context management, tool orchestration, verification UX) compounds as a product asset — a compounding to which Chapter 7 adds one more term: in a knowledge-network world, harness quality also determines how well an organization's knowledge estate is *maintained*, making the harness the custodian of knowledge capital itself — favoring Anthropic's harness-first strategy, whose Cowork-to-enterprise arc plus MCP stewardship plus the Microsoft supply position constitutes the most option-rich single position in the field, with the duality of 6.3.1 as both its upside (distribution) and its tail risk (dependence on a customer-competitor). Second, the labs' enterprise revenue mix (OpenAI past forty percent and rising) means the office layer is becoming the labs' P&L, not a side bet — expect their office-product velocity to remain the field's fastest. Third, independent harness companies (Genspark, Skywork, the next Manus) are structurally attractive acquisition currency for any platform short of agent product — but now carry the geopolitical overhang priced into existence by Beijing's unwinding order: cross-border agent M&A has a new discount factor, and origin-of-technology diligence is no longer optional.

9.3 Adjacent and Niche Positions: Survivor Criteria in the Fourth Bundling War

Stage 3's lesson — single functions get absorbed; category creators survive — carries forward with one amendment: in the agent era the surviving category must be one the bundle cannot cheaply replicate *and* one that agents make more, not less, necessary. Notion qualifies on the strength of its agent conversion and its pricing experiment (its credit-mix trajectory is a leading indicator for the whole industry) — and acquires a second identity under Chapter 7's lens: a linked-block workspace whose pages agents now maintain is a proto-knowledge-network in commercial form, which is simultaneously Notion's deepest claim on the future and its sharpest exposure, since OKF-class open networks commoditize precisely the container Notion sells, leaving its agent and governance layers as the defensible residue. Canva qualifies by expanding from design into a creative operating system with proprietary model assets and a marketing-lifecycle moat. The screen for any other adjacent name reduces to three questions: does it own context agents need, does it own a verification or governance surface, or does it own a protocol position? Positions owning none of the three are, in this framework, bundle food.

9.4 Key Risks

Governance and prompt injection. Agents with folder access, inbox access, and action rights industrialize a new failure class: malicious content in documents manipulating agent behavior, and destructive actions from underspecified instructions. The vendors acknowledge both (Notion ships injection guardrails and run logging; the labs gate high-risk actions behind approvals), but a headline organizational-scale incident would reset enterprise adoption curves and regulatory posture simultaneously. Governance capability is therefore not a feature checkbox but a demand driver — and an argument for the gated estate model.

Geopolitical capture of agent technology. The Manus unwinding establishes the precedent: agent systems are now export-control subject matter. Price it into any thesis involving Chinese-origin harnesses (Skywork included), cross-border agent M&A, or sovereign-AI procurement of office agents.

Regulatory scrutiny of the fourth bundle. The Teams-Slack precedent in Europe primed regulators for exactly the play Microsoft is now running at larger scale with E7. The fourth bundling war is the first one fought under an antitrust regime that has already litigated the third.

The transition-economics trap. For all sellers, the seat-to-consumption migration risks a revenue valley: agents reduce the human-seat count faster than credit consumption replaces it if delegation outpaces monetization. The hybrid bridges of 2026 are designed against this; their credit-mix disclosures are where the trap would first become visible.

Semantic lock-in and the dual-truth problem. The knowledge network resolves format lock-in and reintroduces lock-in one level up: an enterprise whose operating knowledge lives in a vendor's compiled context graph faces switching costs of a new and deeper kind — semantic lock-in, the successor to data lock-in — and the portability of compiled graphs, not of files, becomes the exit question that procurement has not yet learned to ask. Operationally, the same architecture creates a new failure class — the dual-truth problem of Section 7.6: compiled knowledge that drifts from its sources. A stale bundle confidently served to agents is worse than no bundle, and an organizational-scale decision traced to desynchronized knowledge would do for the knowledge-network thesis what a prompt-injection headline would do for agent adoption. Freshness engineering, like governance, is therefore a demand driver in disguise.

Appendices

Appendix A. Timeline of Key Events (2023–2026)

2023 — Microsoft 365 Copilot launches at $30/user/month; Google ships Workspace AI (Duet, later Gemini); Python in Excel introduced: the copilot interlude at full strength.

2024 — Anthropic proposes the Model Context Protocol (MCP); llms.txt proposed (September); NotebookLM's Audio Overviews (September) turn Google's source-grounded notebook into a consumer phenomenon, with the NotebookLM Plus tier following in December.

2025 — *February:* Anthropic releases Claude Code as a research preview (the harness seed). *March:* Manus launches. *May:* Skywork Super Agents launch globally (five-agent "AI Office"). *September:* Notion 3.0 rebuilds around autonomous agents. *October:* Claude for Excel beta; Manus 1.5 (sub-4-minute tasks). *October–November:* Canva Creative Operating System; Affinity free. *November:* Microsoft–Anthropic $30B Azure agreement; Ignite expands Agent Mode across Office; NotebookLM adds slide decks and infographics (on the Nano Banana Pro image model) and integrates Deep Research. *December:* Meta agrees to acquire Manus ($2B+; ~$100M ARR); NotebookLM adds Data Tables with export to Google Sheets.

2026 — *January:* Anthropic launches Claude Cowork (research preview); Anthropic becomes a Microsoft subprocessor; Claude for Excel opens to Pro; China opens Manus export-control review. *February:* OpenAI Frontier; Claude for PowerPoint; Notion 3.3 Custom Agents; Skywork Desktop for Windows; Canva acquires Cavalry/Mango AI. *March:* Microsoft Copilot Wave 3 — Copilot Cowork (Claude-powered) in research preview, Claude in mainline Copilot chat; Google ships the open-source Workspace CLI; NotebookLM adds slide revisions, cinematic Video Overviews, and PPTX export; Copilot Cowork expands to Microsoft's Frontier early-access program (late March). *April:* Google launches Gemini Enterprise Agent Platform (Vertex AI evolution) and announces Workspace Intelligence and the Workspace agent tools at Cloud Next '26; OpenAI Workspace Agents; Canva acquires Simtheory and Ortto, ships agentic AI 2.0; Cowork GA with enterprise controls; China orders the Meta–Manus acquisition unwound (April 27). *May:* Microsoft 365 E7 GA ($99) and Agent 365 add-on ($15); Google Workspace MCP server public preview and API tiering (May 1); Notion Credits billing begins (May 4); OpenAI agent credit pricing begins (May 6); Claude reaches general availability across Excel, PowerPoint, and Word, with Claude for Outlook in public beta (May 7); Google I/O — Gemini 3.5, Gemini Spark (24/7 agent on the Antigravity harness). *June:* Google publishes the Open Knowledge Format, OKF v0.1 (June 12); Work IQ APIs reach general availability (June 16); NotebookLM runs on Gemini 3.5. *July:* Google renames NotebookLM to **Gemini Notebook** (July 16), retaining the standalone product, deepening Gemini-app synchronization with Search AI Mode integration announced, and rolling out per-notebook native code execution.

Appendix B. Format and Protocol Reference Table

Layer	Asset	Status	Strategic reading
Document container	OOXML (.docx/.xlsx/.pptx) — ECMA-376 / ISO 29500	Open standard; Strict vs. Transitional split; rendering unspecified	Neutralized output codec; residual gravity = reference renderer (MS Office)
Fixed rendering	PDF (ISO 32000)	Open; specifies final appearance	The benchmark of format neutrality; the record format of the boundary tier

Layer	Asset	Status	Strategic reading
Cloud-native "format"	Google Docs/Sheets/ Slides internal model	Proprietary; operation-log storage; canvas rendering; no file	Zero format lock-in, maximal service lock-in; "the format is the API"
Agent working tier	Markdown (+ AGENTS.md, SKILL.md, llms.txt)	Open, unownable; Linux Foundation stewardship of agent conventions	Lingua franca of agent work; strategically sterile standardization
Qualitative knowledge tier	OKF — Open Knowledge Format (Markdown + YAML bundles, links as graph)	Open spec v0.1 (Google, June 12, 2026); Knowledge Catalog ingestion; adoption beyond Google unproven	The working tier's qualitative half formalized; interchange container of the knowledge network (7.6–7.7)
Quantitative tier	XLSX / Parquet / SQL-code	Division of labor	Audit surface / data interchange / analytical logic respectively; CSV squeezed out
Interconnect protocols	MCP, A2A	MCP: Anthropic-created, Linux Foundation-run; adopted by MS Agent Store, Google Spark	The .doc war's true successor theater
Estate context APIs	Work IQ APIs; Google Workspace MCP/CLI	GA June 2026; preview/ open-source 2026	Gated-metered vs. open-commoditized postures (Section 5.3)
Commerce protocols	UCP (Google), ACP (OpenAI/Stripe), AP2 (Google-led, industry-collaborative)	Contested, 2026	The most visible live protocol war
Execution layer	Agent sandboxes/runtimes (Agent Platform workspaces, lab environments, Claw)	Productizing	Where container lock-in relocates under the decomposition thesis

Appendix C. Glossary

Actuator — the role of a legacy application when operated by an agent rather than a human (Phase A). **Agent harness** — the orchestration system around a model: planning, tool use, context management, verification, recovery; the era's scarce engineered asset. **Boundary tier / working tier** — the two-tier format regime: human-institutional artifacts (DOCX/XLSX/PPTX/PDF) versus agent-native representations (MD/JSON/Parquet/code). **Context graph** — the semantic model of an organization's people, artifacts, and activity; the by-product of office usage, productized as Work IQ / Workspace Intelligence. **Decomposition thesis** — the unbundling of fused office containers into data + code + web rendering. **Fourth bundling war** — Microsoft's current E7/Agent 365 campaign, successor to the Office, Teams, and Copilot bundles. **Knowledge capital** — the asset a knowledge network constitutes: the refined, compounding, agent-maintained form of the enterprise's revalued qualitative and quantitative knowledge. **Knowledge network** — an organization's knowledge as a curated, linked, machine-executable estate — concepts, typed data, and declared logic — from which artifacts are compiled on demand; this report's projected end-state for the office category. **MCP** — Model Context Protocol; the open agent-tool interconnect created by Anthropic, stewarded by the Linux Foundation. **Metric concept** — business logic (a KPI definition or formula) expressed as a governed, versioned knowledge concept binding a declared definition to a typed dataset; the semantic layer in file form. **OKF (Open Knowledge Format)** — Google's open v0.1 specification (June 2026) representing organizational knowledge as directories of Markdown files with YAML frontmatter, ordinary links forming a knowledge graph; the formalization of the LLM-wiki pattern. **Reference renderer** — the application whose rendering behavior defines de facto compatibility for a structurally specified format (MS Office for OOXML). **Source-first** — the office paradigm in which a curated corpus is primary and artifacts are generated renderings of it (Gemini Notebook), inverting the triad's blank-artifact model. **Supplier-competitor duality** — the era's signature relationship: one firm simultaneously powering and invading another's product (Anthropic–Microsoft). **Triad** — Document/Spreadsheet/Presentation: the three institutionalized cognitive axes of knowledge work. **Work IQ / Workspace Intelligence** — Microsoft's and Google's productized organizational context layers.

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